Bioorganic & Medicinal Chemistry Letters 30 (2020) 126949

Contents lists available at ScienceDirect

Bioorganic & Medicinal Chemistry Letters

journal homepage: www.elsevier.com/locate/bmcl

Orally bioavailable amine-linked macrocyclic inhibitors of factor XIa

Tianan Fang[a,*], James R. Corte[a,*], Paul J. Gilligan[a], Yoon Jeon[a], Honey Osuna[a], Karen A. Rossi[a], Joseph E. Myers Jr.[b], Steven Sheriff[b], Zhen Lou[c], Joanna J. Zheng[c], Timothy W. Harper[c], Jeffrey M. Bozarth[c], Yiming Wu[c], Joseph M. Luettgen[c], Dietmar A. Seiffert[c], Ruth R. Wexler[a], Patrick Y.S. Lam[a]

[a] Bristol-Myers Squibb Company, Research and Development, 350 Carter Road, Hopewell, NJ 08540, United States
[b] Bristol-Myers Squibb Company, Research and Development, US Rt. 206 & Province Line Road, Princeton, NJ 08540, United States
[c] Bristol-Myers Squibb Company, Research and Development, 311 Pennington-Rocky Hill Road, Pennington, NJ 08543-2130, United States

ARTICLE INFO

Keywords:
Factor Xia
FXIa
Activated partial thromboplastin time
aPTT
Thrombosis

ABSTRACT

The discovery of orally bioavailable FXIa inhibitors has been a challenge. Herein, we describe our efforts to address this challenge by optimization of our imidazole-based macrocyclic series. Our optimization strategy focused on modifications to the P2 prime, macrocyclic amide linker, and the imidazole scaffold. Replacing the amide of the macrocyclic linker with amide isosteres led to the discovery of substituted amine linkers which not only maintained FXIa binding affinity but also improved oral exposure in rats. Combining the optimized macrocyclic amine linker with a pyridine scaffold afforded compounds **23** and **24** that were orally bioavailable, single-digit nanomolar FXIa inhibitors with excellent selectivity against relevant blood coagulation enzymes.

Cardiovascular diseases (CVDs), the leading cause of death worldwide, accounted for about 32% of the 55.9 million global deaths in 2017.[1] The underlying pathophysiology of CVDs often involves the formation of a thrombus that rapidly slows or stops blood flow potentially leading to heart attack or stroke. The direct thrombin inhibitor dabigatran and the factor Xa inhibitors (rivaroxaban, apixaban, edoxaban, and betrixaban) are effective for the treatment or prevention of deep vein thrombosis (DVT), pulmonary embolism (PE) or stroke arising from nonvalvular atrial fibrillation (NVAF).[2,3] However, data from clinical trials and post-marketing data indicates that bleeding remains a safety concern, especially in patients with increased bleeding risk.[4] Thus, a need exists for an oral antithrombotic with a wider therapeutic index with respect to the bleeding safety profile. Factor XIa (FXIa), the activated form of the zymogen FXI, plays a major role in the amplification pathway of the coagulation cascade but not in hemostasis.[5] Inhibition of FXIa has the potential of providing an effective antithrombotic agent with limited effects on hemostasis. Preclinical studies have shown that peptide and small molecule FXIa inhibitors prevent thrombosis with less bleeding liability.[6,7]

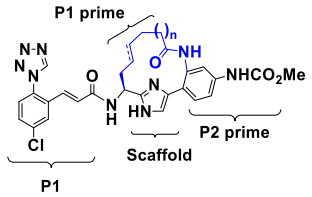
Several small molecule irreversible,[6,8] reversible,[7,9,10] and allosteric[11] FXIa inhibitors have been reported in the literature. Bristol-Myers Squibb has reported on the discovery of a novel series of 12- and 13-membered imidazole-based macrocyclic FXIa inhibitors **1** and **2** (Table 1).[12] These macrocycles with an amide-linker displayed high FXIa binding affinity (FXIa Ki $\leq$1 nM)[13] and are potent in the activated partial thromboplastin (aPTT) clotting assay (aPTT $EC_{1.5x} \leq 0.30$ μM);[14] however, these compounds were not orally bioavailable ($\%F_{rat} < 1$). We hypothesized that high polar surface area (PSA = 169 Å^2 for **1** and **2**) may contribute to poor permeability [Caco-2 AB/BA (nm/s): < 15/67 for **1** and < 15/58 for **2**] and lack of oral bioavailability.[15] Our initial optimization strategy for improving oral bioavailability by replacing the chlorophenyltetrazole P1 with less polar groups to reduce PSA led to the discovery of novel P1 groups that proved to be successful for improving oral bioavailability.[16] Described herein is an alternative strategy in which the chlorophenyltetrazole P1 was kept fixed and the P2 prime, macrocyclic amide linker, and scaffold regions of the macrocycle were modified with the goal of improving the oral bioavailability. In this communication, we describe SAR in these three regions culminating in the discovery of a macrocyclic amine linker which maintained FXIa binding affinity and improved oral bioavailability for both the imidazole- and pyridine-based macrocyclic series.

First, we explored the P2 prime region of the macrocycle by removing the carbamate moiety to reduce PSA (Table 2). Based on X-ray crystallography, the carbamate moiety is engaged in several H-bonding interactions with FXIa (the NH of the carbamate forms an H-bond to

* Corresponding authors.
E-mail addresses: tianan.fang@bms.com (T. Fang), james.corte@bms.com (J.R. Corte).

https://doi.org/10.1016/j.bmcl.2020.126949
Received 15 November 2019; Received in revised form 23 December 2019; Accepted 1 January 2020
Available online 07 January 2020
0960-894X/ © 2020 Elsevier Ltd. All rights reserved.

Table 1
12- and 13-membered macrocyclic FXIa inhibitors (the macrocyclic amide linker is highlighted in blue).



Compd #	n (ring size)	Saturation/unsaturation	FXIa Ki (nM)[13]	aPTT $EC_{1.5x}$ (μM)[14]
1	0 (12)	saturated	1.0	0.30
2	1 (13)	E-alkene	0.16	0.27

His40 and the carbonyl of the carbamate forms an H-bond to Ile151 via a water molecule) and it was not clear how the loss of these interactions would affect FXIa affinity.[12] Removing the carbamate moiety in 12-membered macrocycle **3** lowered PSA (130 $Å^2$) but also led to a significant loss in FXIa affinity (48-fold; FXIa Ki = 48 nM), aPTT potency (> 100 fold; $EC_{1.5x}$ > 40 μM) and metabolic stability [15-fold; human liver microsome (HLM) $t_{1/2}$ = 6 min]. A similar loss in FXIa binding affinity (32-fold) and metabolic stability was observed when the carbamate was removed in the 13-membered macrocycle **4**; however, a single-digit nanomolar FXIa inhibitor (FXIa Ki = 5.1 nM) with good aPTT potency ($EC_{1.5x}$ = 2.9 μM) was realized. Encouraged by this result in the 13-membered macrocycle, additional substituents were explored at the 4- and 5-positions of the P2 prime phenyl group with the goal of improving metabolic stability. Whereas introduction of a fluoro at C5 (**5**) resulted in a slight loss in FXIa affinity, C4-F (**6**) was equipotent with **4**. Unfortunately, neither **5** nor **6** improved HLM stability. Compound **8**, with a carboxylic acid group at C4, was the only analog that maintained FXIa affinity and improved metabolic stability (HLM $t_{1/2}$ = 94 min), but unfortunately PSA (168 $Å^2$) was high. Since efforts to reduce PSA in the P2 prime region led to a loss in both FXIa affinity and metabolic stability, we turned our attention to modifying the amide moiety in the macrocyclic linker.

During our initial exploration of the imidazole-based macrocyclic series, the macrocycles with an amide linker were identified as giving the most potent FXIa inhibitors.[12] It was demonstrated by X-ray crystallography that the NH of the amide moiety forms a key H-bond with Leu41 (2.8 Å) which contributed significantly to FXIa binding affinity but proved to be deleterious for oral bioavailability.[12] The literature suggests that amide isosteres may be a means of modulating polarity and oral bioavailability.[17] We were particularly interested in replacing the amide moiety with a suitable isostere that had the potential to maintain the key H-bonding interaction with Leu41 such as a trifluoroethylamine [–$CH(CF_3)$NH–] group. The trifluoroethylamine unit has been adopted as an effective amide isostere in drug design since it is a non-basic amine with excellent H-bond donating ability.[18,19] However, it was not clear whether the trifluoroethylamine moiety, once it was incorporated into the linker, would be able to adopt the desired conformation to maintain the key H-bond with Leu41. Moreover, an unsubstituted macrocycle linked by an amine was prepared earlier in a 13-membered macrocycle and it showed a significant loss in FXIa affinity.[12]

Due to the synthetic route, two diastereomers were produced by incorporating the trifluoroethylamine group into the macrocyclic backbone. In the 12-membered macrocycle, both diastereomers (*S*)-CF_3 **9** and (*R*)-CF_3 **10** (Table 3) exhibited a desired, lower PSA value (152 $Å^2$ vs 169 $Å^2$ for amide **1**). Diastereomer (*S*)-CF_3 **9** exhibited a significant loss (370-fold) in FXIa affinity, while diastereomer (*R*)-CF_3 **10** showed similar FXIa binding affinity (FXIa Ki = 1.1 nM) compared to amide **1**, however a 9-fold loss in aPTT clotting activity ($EC_{1.5x}$ = 2.6 μM) was seen. The trifluoroethylamine proved to be a good replacement for the amide moiety in the 12-membered macrocycle but not the 13-membered macrocycle **14**, where a 330-fold loss in FXIa affinity (FXIa Ki = 53 nM) was observed. Since 12-membered macrocycles with an amine linker were more potent, additional substitutions were explored. Diastereomer (*R*)-CHF_2 **11** was equipotent to (*R*)-CF_3 **10**. When the fluoro alkyl groups were replaced with either CH_3 as in **12** or hydrogen as in **13**, a loss in both FXIa affinity (> 6-fold) and metabolic stability was observed. The difference in FXIa binding affinity can be rationalized based on the electron withdrawing ability of the R-substituent. Specifically, the stronger electron withdrawing ability of the fluoro alkyl substituents in **10** and **11** makes the adjacent NH a better H-bond donor for Leu41 compared to either the methyl or hydrogen substituent in **12** and **13**.

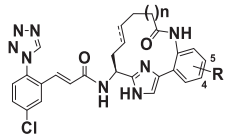
An X-ray crystal structure of **10** bound to human FXIa active site (2.2 Å resolution, Fig. 1) was obtained.[20] The compound occupies the S1, S1 prime, and the S2 prime binding pockets. The absolute stereochemistry of the trifluoroethylamine moiety was determined to be the *R*-configuration. An overlay of trifluoroethylamine linker **10** and amide linker **1** is shown in Fig. 2 and highlights the interaction of the trifluoroethylamine moiety. As postulated in the design, the trifluoroethylamine moiety maintains the key H-bond (3.0 Å) with Leu41 and mimics the amide interaction. Interestingly, the X-ray crystal structure also revealed that the CF_3 group, known to be a weak H-bond acceptor,[21] is within hydrogen bonding distance (3.0 Å) to Arg39. This weak H-bonding interaction may also be contributing to the FXIa binding affinity of the fluoro alkyl analogs **10** and **11**, but not **12** or **13**.

In order to explore this potential interaction between the CF_3 group and Arg39, the CF_3 moiety was replaced with other potential H-bond acceptors (Table 4). We focused on 12-membered macrocycles since the 13-membered macrocycle with the trifluoroethylamine linker was much less potent. Replacing the CF_3 with methyl ester **15** resulted in a 2-fold increase in FXIa affinity (FXIa Ki = 0.46 nM) and a 5-fold improvement in aPTT clotting potency ($EC_{1.5x}$ = 0.55 μM). The ethyl ester **16** was similar in potency to methyl ester **15**. These ester analogs had high PSA values (178 $Å^2$) and poor metabolic stability (HLM $t_{1/2}$ ≤10 min). Acid **17** was similar in potency to the corresponding ester analogs and exhibited improved metabolic stability (HLM $t_{1/2}$ = 70 min). Amide **18** and hydroxymethyl **19** analogs displayed good FXIa affinity and aPTT potency but also suffered from poor metabolic stability. Homologation of methyl ester **20** led to a 4- to 5-fold loss in FXIa affinity. The NH moiety in the macrocyclic linker proved to be the major contributor to FXIa affinity, as removal of the NH led to a 50-fold loss in FXIa binding affinity for the all-carbon linker **21**.[22] In general, the imidazole-based macrocycles containing the substituted amine linkers in Table 4 had high PSA values (≥172 $Å^2$) and, aside from acid **17**, suffered from poor metabolic stability (HLM $t_{1/2}$ ≤ 14 min).

X-ray crystal structures of methyl ester **15** (2.12 Å resolution, Fig. 3) and ethyl ester **16** (2.07 Å resolution, Fig. 4A) bound to human FXIa active site were obtained.[23] The carbonyl of the methyl ester in **15** formed an H-bond to Arg39 (2.8 Å). The ether oxygen forms an H-bond to the Cys58 backbone carbonyl via two water molecules. Interestingly, the ethyl ester in **16** does not form an H-bond to Arg39. Rather, the ethyl ester is flipped so that the ethyl portion forms a van der Waals contact with a small pocket created by residues 39–41 and the carbonyl makes an H-bond to Cys58 carbonyl through two water molecules (Fig. 4B).

Previous work from our lab on the macrocyclic series showed that the imidazole scaffold can be replaced with a pyridine ring system and maintain FXIa binding affinity.[16c] As a result, the optimized amine linkers were combined with a pyridine scaffold (Table 5). The trifluoroethylamine linker in pyridine macrocycle **22** led to a 14-fold loss in FXIa affinity (FXIa Ki = 16 nM) and a 5-fold loss in aPTT activity ($EC_{1.5x}$ = 13 μM) when compared to imidazole-based analog **10**.

Table 2
P2 prime SAR in the 12- and 13-membered imidazole macrocycles.



Compd #	n (ring size)	Saturation/unsaturation	R	FXIa Ki (nM)[13]	aPTT $EC_{1.5x}$ (μM)[14]	PSA[a] ($Å^2$)	HLM[b] $t_{1/2}$ (min)
1	0 (12)	saturated	5-$NHCO_2Me$	1.0	0.30	169	90
3	0 (12)	saturated	H	48	> 40	130	6
2	1 (13)	E-alkene	5-$NHCO_2Me$	0.16	0.27	169	41
4	1 (13)	E-alkene	H	5.1	2.9	130	3
5	1 (13)	E-alkene	5-F	12	16	130	3
6	1 (13)	E-alkene	4-F	6	2.5	130	2
7	1 (13)	E-alkene	4-CO_2Me	245	NT[c]	157	6
8	1 (13)	E-alkene	4-CO_2H	3.6	1.7	168	94

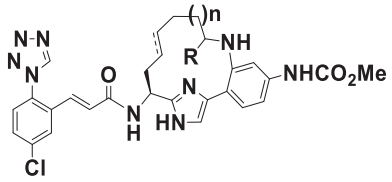
[a] PSA = polar surface area.
[b] HLM = human liver microsome stability.
[c] NT = not tested.

Replacing the CF_3 moiety with either CO_2Me **23** or CO_2Et **24** improved FXIa binding affinity, resulting in single-digit nanomolar inhibitors (FXIa Ki = 3.4 nM and 2.5 nM, respectively) with acceptable aPTT clotting activity. While less potent than the corresponding imidazole-based macrocycles, these pyridine-based macrocycles containing the amine-linker had lower PSA values (162 $Å^2$) and improved metabolic stabilities (HLM $t_{1/2}$ = 111 min. for **23** and 30 min. for **24**) compared to their imidazole analogs. Acid **25** was equipotent to the corresponding ester analogs, but further increased the PSA value (173 $Å^2$).

An X-ray crystal structure of ethyl ester **24**[24] bound to human FXIa active site (1.89 Å resolution, Supplementary Fig. S1) was obtained and binds like imidazole macrocycle **16**, where the ethyl portion of the ester is interacting with the small pocket and not engaging Arg39 via an H-bond.

Several compounds were evaluated in a rat pharmacokinetic (PK) model, in an *N*-in-1 format (Table 6).[25] The 12-membered imidazole macrocycle **1** exhibited a low clearance, short half-life, low volume of distribution, and was not orally bioavailable. Removing the carbamate from the P2 prime region in **3** resulted in a significant increase in clearance and no oral exposure, in spite of a lower PSA (130 $Å^2$); the efflux ratio was high (Caco-2 AB/BA (nm/s) < 15/205, ER > 13).[26] Replacing the amide moiety in **1** with the trifluoroethylamine group in **10** not only maintained low clearance but improved oral exposure (**10**, AUC = 307 nM*h versus **1**, AUC = 24 nM*h) leading to an oral bioavailability of 9%. Methyl ester, **15**, was found to be a very low clearance compound and was orally bioavailable (%F = 8) despite high PSA (178 $Å^2$) and poor permeability (Caco-2 AB/BA (nm/s) < 15/103). While ethyl ester **16** displayed a similar PK profile to **15** and was orally bioavailable (%F = 7), acid **17** showed only minimal exposure in rat. Replacing the imidazole scaffold of **15** with a pyridine in **23** further improved oral bioavailability (%F = 23) while maintaining a low clearance, short half-life, and low volume of distribution, despite high

Table 3
Amine linker SAR in 12- and 13-membered imidazole macrocycles.



Compd #	n (ring size)	Saturation/unsaturation	R	FXIa Ki (nM)[13]	aPTT $EC_{1.5x}$ (μM)[14]	PSA[a] ($Å^2$)	HLM[b] $t_{1/2}$ (min)
1	0 (12)	saturated	–	1.0	0.30	169	90
9	0 (12)	saturated	(*S*)-CF_3[c]	370	NT[d]	152	NT[d]
10	0 (12)	saturated	(*R*)-CF_3[c]	1.1	2.6	152	54
11	0 (12)	saturated	(*R*)-CHF_2[c]	1.3	2.0	152	25
12	0 (12)	saturated	(*S*)-CH_3[e]	8.3	5.2	152	7
13	0 (12)	saturated	H	16	6.2	152	6
2	1 (13)	E-alkene	–	0.16	0.27	169	41
14	1 (13)	E-alkene	(*R*)-CF_3[c]	53	> 40	152	29

[a] PSA = polar surface area.
[b] HLM = human liver microsome stability.
[c] The absolute stereochemistry was assigned based on an X-ray co-crystal of **10** with FXIa (Fig. 1).
[d] NT = not tested.
[e] The configuration of the stereocenter is the same as for (*R*)-CF_3 **10** and (*R*)-CHF_2 **11**, however the *R* designation changed to *S* based on the CH_3 group which altered the prioritization in the Cahn-Ingold-Prelog system.

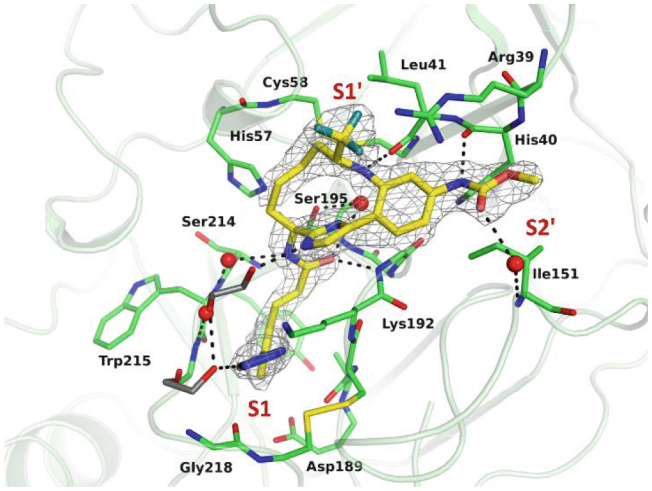


Fig. 1. X-ray crystal structure of **10** bound to Factor XIa with omit electron density contoured at 3 rmsd (gray mesh). The red spheres depict water molecules, the dotted lines depict hydrogen bonds, and ethylene diol is an artifact of the flash-cooling procedure. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

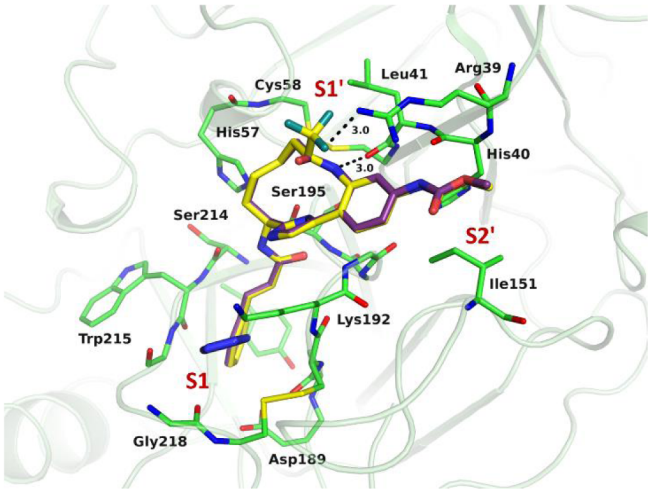


Fig. 2. Overlay of X-ray crystal structures of **1** (purple) and **10** (yellow) in Factor XIa. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

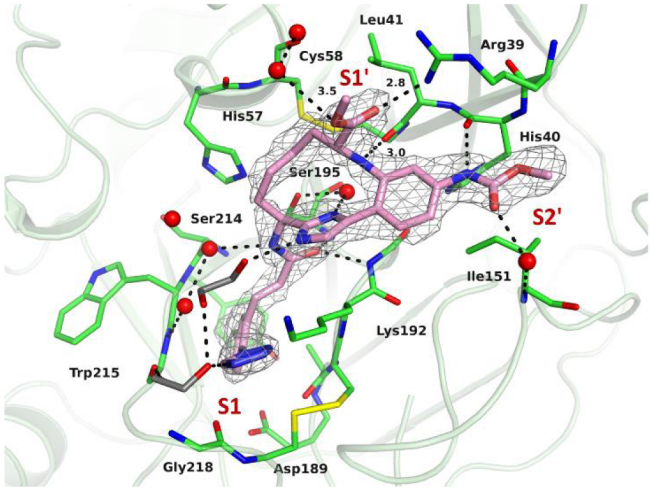
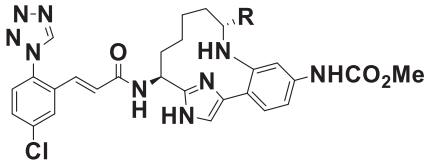


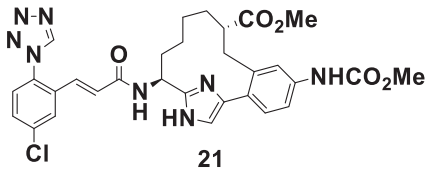
Fig. 3. X-ray crystal structure of **15** bound to Factor XIa with omit electron density contoured at 3 rmsd (gray mesh). The red spheres depict water molecules, the dotted lines depict hydrogen bonds, and ethylene diol is an artifact of the flash-cooling procedure. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

PSA (162 Å^2). The ethyl ester **24** exhibited a low clearance, a longer half-life, and a higher volume of distribution and good oral bioavailability (%F = 41). In addition to improved oral bioavailability for the amine-linked macrocycles, imidazole **16** and pyridine **24** were > 1,000-fold selective against a panel of relevant serine proteases (Table S1 in Supplementary Data Section).

All macrocycles were prepared using ring-closing olefin metathesis (RCM) as the key macrocyclization step.[27] The synthesis of macrocyclic analogs in which the P2 prime region was modified (Table 2) follows a synthetic sequence previously described by Corte et al.[12] that is included in the Supplementary Data (Schemes S1 and S2). The synthesis of representative imidazole macrocycles with amine-linker **10**, **15** and **17** (Tables 3 and 4) is described in Scheme 1. Condensation of aniline **1a**[12] with 1-ethoxy-2,2,2-trifluoroethanol in EtOH at elevated temperature gave the corresponding aminal, which was then reacted with allyl magnesium bromide which provided trifluoroethylamine **1b** as a 1:1 mixture of diastereomers.[28] Pretreatment of **1b** with *p*-toluenesulfonic acid to form the imidazolium ion followed by macrocyclization via RCM using Grubbs II catalyst gave imidazole macrocycle **1c** as a mixture of diastereomers.[29] Hydrogenation of **1c** over palladium on carbon in the presence of TFA, followed by separation of diastereomers

Table 4
Substituted amine linker SAR in the 12-membered imidazole macrocycle.





21

Compd #	R	FXIa Ki (nM)[13]	aPTT $EC_{1.5x}$ (μM)[14]	PSA[a] (Å^2)	HLM[b] $t_{1/2}$ (min)
15	CO_2Me	0.46	0.55	178	10
16	CO_2Et	0.22	0.86	178	7
17	CO_2H	0.67	0.87	189	70
18	$C(O)NMe_2$	0.31	0.36	172	14
19	CH_2OH	0.61	0.54	172	10
20	CH_2CO_2Me	2.0	1.1	178	4
21	–	24	25	166	8

[a] PSA = polar surface area.
[b] HLM = human liver microsome stability.

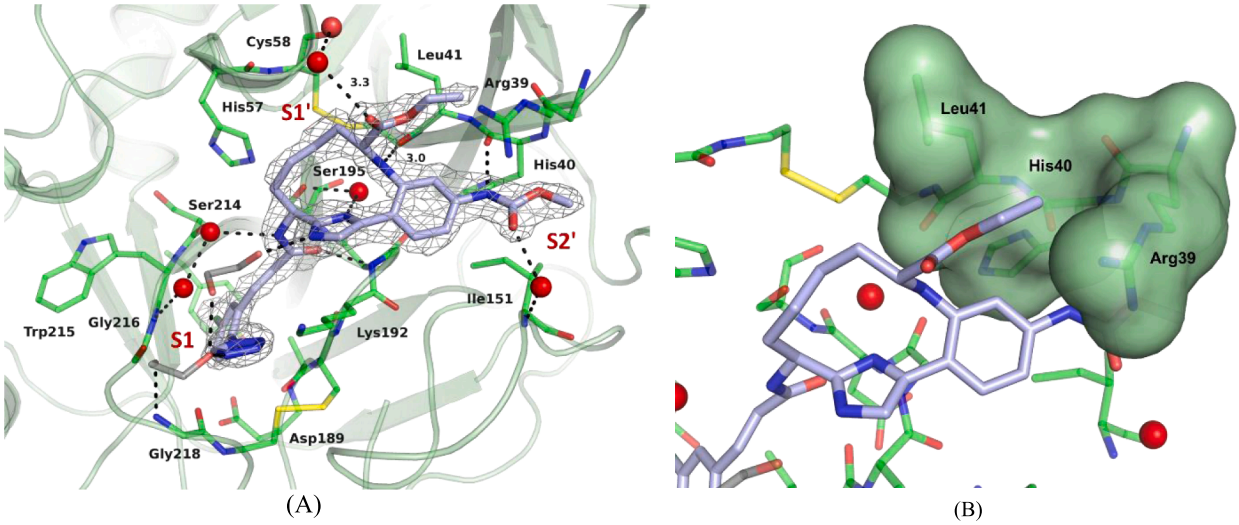
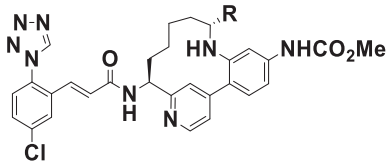


Fig. 4. (A) X-ray crystal structure of **16** bound to Factor XIa with omit electron density contoured at 3 rmsd (gray mesh). The red spheres depict water molecules, the dotted lines depict hydrogen bonds, and ethylene diol is an artifact of the flash-cooling procedure. (B). van der Waals contact between the ethyl portion of the ester and small pocket formed by residues 39–41. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Substituted amine linker SAR in the 12-membered pyridine macrocycle.



Compd #	R	FXIa Ki (nM)[13]	aPTT $EC_{1.5x}$ (μM)[14]	PSA[a] (Å^2)	HLM[b] $t_{1/2}$ (min)
22	CF_3	16	13	136	95
23	CO_2Me	3.4	3.4	162	111
24	CO_2Et	2.5	6.1	162	30
25	CO_2H	2.6	7.1	173	99

[a] PSA = polar surface area.
[b] HLM = human liver microsome stability.

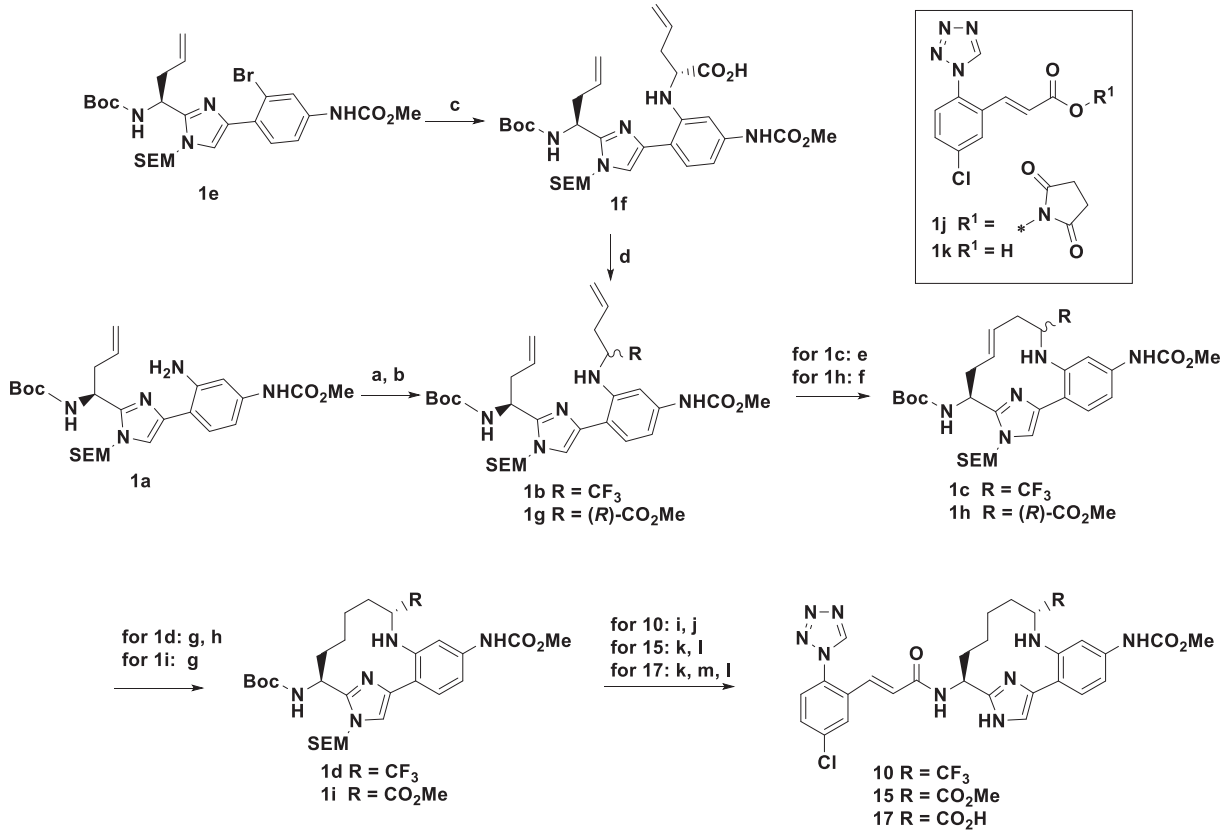
by preparatory reverse phase hplc, afforded macrocycle **1d**. Global deprotection with aqueous 5 N HCl provided the corresponding amine, which was coupled with acid **1k** to give **10**. Alternatively Ullmann coupling of bromide **1e**[12] with (*R*)-2-aminopent-4-enoic acid provided acid **1f** which was converted to methyl ester **1g** under basic conditions with iodomethane. Macrocyclization of **1g** via RCM using Grubbs II catalyst with microwave heating gave macrocycle **1h**. Hydrogenation of alkene in **1h** yielded **1i**. Deprotection of **1i** as described above followed by coupling with activated carboxylic ester **1j** provided methyl ester **15**. Compound **1i** was converted to acid **17** in a three step sequence which involved deprotection of the Boc and SEM groups, hydrolysis of the methyl ester under basic conditions, and finally coupling with **1j**. The synthesis for all-carbon imidazole macrocycle **21** was carried out by an alternative sequence and can be found in Scheme S3 (Supplementary Data).

The syntheses of representative pyridine-based macrocycles with amine-linker **22**, **24** and **25** (Table 5) is described in Scheme 2. Similar to the imidazole synthesis, aniline **2a**[16c] was condensed with 1-ethoxy-2,2,2-trifluoroethanol, followed by addition of allyl magnesium

Table 6
Pharmacokinetic profile for selected 12-membered macrocyclic compounds.

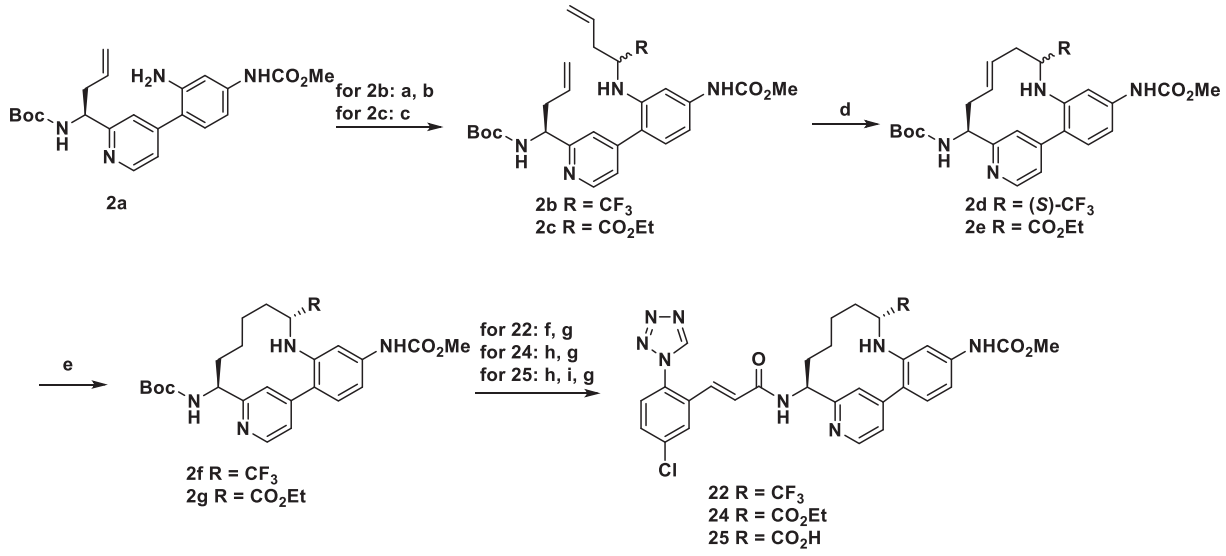
Entry	Cl (mL/min/kg)	$t_{1/2}$ (h)	Vd_{ss} (L/kg)	F (%)	AUC[a] (nM*h)	PSA[d] (Å^2)	Dose iv/po (mpk)	PB[c] rat (%free)	RLM[g] $t_{1/2}$ (min)	Caco-2 AB/BA (nm/*sec*)
1	10	1.0	0.3	< 1	24	169	1.13/2.27	1.6	76	< 15/67[h]
3	53	0.5	1.2	< 1	1	130	0.31/0.61	NT[f]	NT[f]	< 15/205
10	12	1.4	0.4	9	307[c]	152	0.72/1.44	0.2	14	NT[f]
15	3.0	1.8	0.3	8	487[b]	178	0.53/1.06	< 0.2	48	< 15/103
16	2.8	1.5	0.2	7	478[b]	178	0.50/1.0	< 0.2	34	< 15/133[h]
17	7.7	1.0	0.33	1	14[b]	189	0.49/0.98	1.4	75	< 15/ < 15
23	5.2	1.2	0.53	23	889[b]	162	0.52/1.04	NT[f]	87	< 15/213
24	4.9	3.6	1.3	41	1294	162	0.50/1.0	0.8	59	< 15/101[h]

Compounds were dosed in rat in an *N*-in-1 format. Area under the curve (AUC) values were from the po dose.
[a] Vehicle for both iv and po: 70% polyethylene glycol 400; 20% water; 10% ethanol.
[b] Vehicle for both iv and po: 70% propylene glycol; 20% water; 10% ethanol.
[c] Compound was dosed in rat in a discrete format.
[d] PSA = polar surface area.
[e] PB = protein binding.
[f] NT = not tested.
[g] RLM = rat liver microsome stability.
[h] Poor recovery in the Caco-2 permeability assay.



Scheme 1. Reagents and conditions: (a) 1-ethoxy-2,2,2-trifluoroethanol, EtOH, 120 °C, 86%; (b) allyl magnesium bromide, THF, 0 °C to rt, 50%; (c) (*R*)-2-aminopent-4-enoic acid, CuI, K_2CO_3, DMSO, 90 °C, 96%; (d) MeI, K_2CO_3, DMF, 71%; (e) pTsOH, DCM, reflux, then 40 mol% Grubbs II catalyst, 63%; (f) 40 mol% Grubbs II catalyst, DCE, 120 °C (microwave), 30%; (g) H_2, 10% Pd/C, TFA (2 eq.), EtOAc, for **1i** 52%; (h) preparatory HPLC to separate diastereomers, 9%; (i) 5 N aq. HCl, hydroxylamine, 75 °C, 49%; (j) **1k**, EDC, HOBt, TEA, DMF, 50%; (k) 4 N HCl, cysteine, dioxane, 75 °C, 81%; (l) **1j**, DIPEA, DMF, 37–42%; (m) 1 N NaOH, MeOH, 55 °C, 92%.

bromide which gave trifluoroethylamine **2b** as a mixture of diastereomers. Pyridine **2b** was treated with *p*-toluenesulfonic acid to form the pyridinium ion and then cyclized via RCM using Grubbs II catalyst to give, following separation of the diastereomers by normal phase chromatography, pyridine macrocycle **2d**. Hydrogenation of **2d** provided saturated macrocycle **2f**. The Boc group in **2f** was removed with TFA, and the corresponding amine was coupled with **1j** to give **22**. Ester **24** and acid **25** analogs were prepared via a slightly modified sequence.



Scheme 2. Reagents and conditions: (a) 1-ethoxy-2,2,2-trifluoroethanol, pTsOH, EtOH, 120 °C, 54%; (b) allyl magnesium bromide, THF, −20 °C, 80%; (c) ethyl 2-oxoacetate, maleic acid, then allyltributyltin, ACN, 22%; (d) pTsOH, DCM, reflux, then 40 mol% Grubbs II catalyst, for **2d**, 18% after separation of diastereomers by normal phase chromatography, for **2e**, 60% as a mixture of diastereomers; (e) H_2, 10% Pd/C, TFA, EtOH, for **2f** 80%, for **2g** 20% after separation of diastereomers by normal phase chromatography; (f) 30% TFA in DCM, 87%; (g) **1j**, DIPEA, DMF, 65–83%; (h) 4 N HCl in dioxane, 100%; (i) 1 N NaOH, MeOH, 55 °C, 100%.

Condensation of **2a** with ethyl 2-oxoacetate, followed by addition of allyl tributyltin afforded amino ester **2c** as a mixture of diastereomers. Amino ester **2c** was subjected to RCM conditions which gave unsaturated macrocycle **2e**. The alkene in **2e** was reduced as previously described to give, following separation of diastereomers by normal phase chromatography, saturated macrocycle **2g**. The Boc group in **2g** was removed with 4 N HCl in dioxane, and the corresponding amine was coupled with **1j** to give **24**. Compound **2g** was converted to acid **25** via a three step sequence as was previously described for acid **17**.

In conclusion, our efforts to improve the oral bioavailability of the macrocyclic FXIa series focused on modifying the P2 prime, macrocyclic amide linker, and imidazole regions. Removing the methyl carbamate P2 prime resulted in a significant loss in FXIa affinity and did not improve the oral bioavailability. The oral bioavailability of the series was significantly improved by exploring SAR in the macrocyclic linker and imidazole regions. Specifically, replacing the amide moiety in the macrocyclic linker with amide isosteres led to the discovery of the imidazole macrocycle with the amine linker which improved the oral exposure in rat (**10**, **15** and **16**). Finally, replacing the imidazole scaffold with the pyridine ring further improved oral bioavailability and afforded compound **24**, a single-digit nanomolar FXIa inhibitors with excellent selectivity against relevant blood coagulation enzymes. Further improvements in potency for the pyridine-based macrocycles will be described in subsequent work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank William R. Ewing, Mimi L. Quan, and Joanne M. Smallheer for critical review of this manuscript; the Department of Discovery Synthesis; Dr. Anuradha Gupta, Dr. Manivel Pitchai, and Sankara Narayanan for the synthesis of intermediates at BMS Biocon Research Center; the Department of Lead Discovery & Optimization; and Mojgan Abousleiman for in vitro work. Some of this research used resources of the Advanced Photon Source, a U.S. Department of Energy (DOE) Office of Science User Facility operated for the DOE Office of Science by Argonne National Laboratory under Contract No. DE-AC02-06CH11357. Use of the IMCA-CAT beamline 17-ID at the Advanced Photon Source was supported by the companies of the Industrial Macromolecular Crystallography Association through a contract with Hauptman-Woodward Medical Research Institute.

Appendix A. Supplementary data

Supplementary data (Table S1 listing the human serine protease selectivity profile; Scheme S1 and S2 describing the synthesis of the analogs in Table 2. Scheme S3 describing the synthesis of the compound **21**; X-ray crystal structure of compound **24**; Crystallographic data and refinement statistics for X-ray structures of the FXIa complexes with compounds **10**, **15**, **16**, and **24**) to this article can be found online at https://doi.org/10.1016/j.bmcl.2020.126949.

References

1. GBD Collaborators. *Lancet.* 2018;392:1736.
2. Yeh CH, Hogg K, Weitz JI. *Arterioscler Thromb Vasc Biol.* 2015;35:1056.
3. Cohen AT, Harrington R, Goldhaber SZ, et al. *Am Heart J.* 2014;167:335.
4. Connolly DJ, Ezekowitz MD, Yusuf S, et al. *N Engl J Med.* 2009;361:1139.
5. Bane CE, Gailani D. *Drug Discovery Today.* 2014;19(9):1454.
6. (a) Lin J, Deng H, Jin L, et al. *J Med Chem.* 2006;49:7781
   (b) Schumacher WA, Seiler SE, Steinbacher TE, et al. *Eur J Pharmacol.* 2007;570:167
   (c) Wong PC, Crain EJ, Watson CA, Schumacher WAJ. *Thromb Thrombolysis.* 2011;32:129.
7. (a) Quan ML, Wong PC, Wang C, et al. *J Med Chem.* 2014;57:955
   (b) Wong PC, Quan ML, Watson CA, et al. *Thromb Thrombolysis.* 2015;40:416
   (c) Hu Z, Wong PC, Gilligan PJ, et al. *ACS Med Chem Lett.* 2015;6:590
   (d) Wang X, Kurowski S, Wu W, et al. *J Pharmacol Exp Ther.* 2017;360:476
   (e) Pinto DJP, Orwat MJ, Smith LM, et al. *J Med Chem.* 2017;60:9703.
8. (a) Buchanan MS, Carroll AR, Wessling D, et al. *J Med Chem.* 2008;51:3583
   (b) Deng H, Bannister TD, Jin L, et al. *Bioorg Med Chem Lett.* 2006;16:3049.
9. (a) Hangeland JJ, Friends TJ, Rossi KA, et al. *J Med Chem.* 2014;57:9915
   (b) Pinto DJ, Smallheer JM, Corte JR, et al. *Bioorg Med Chem Lett.* 2015;25:1635.
10. (a) Corte JR, Fang T, Hangeland JJ, et al. *Bioorg Med Chem Lett.* 2015;25:925
    (b) Corte JR, Fang T, Pinto DJP, et al. *Bioorg Med Chem.* 2016;24:2257.
11. (a) Al-Horani RA, Gailani D, Desai UR. *Thromb Res.* 2015:379
    (b) Argade MD, Mehta AY, Sarkar A, Desai UR. *J Med Chem.* 2014;57:3559
    (c) Al-Horani RA, Ponnusamy P, Mehta AY, Gailani D, Desai UR. *J Med Chem.* 2013;56:867.
12. Corte JR, Fang T, Osuna H, et al. *J Med Chem.* 2017;60:1060.
13. Ki values were obtained from purified human enzyme at 37 °C and were averaged from multiple determinations ($n \geq 2$), as described in Refs. 7a,12.
14. aPTT (activated partial thromboplastin time) in vitro clotting assay was performed in human plasma as described in Refs. 7a,12. The reported EC1.5x values are the FXIa inhibitor plasma concentrations which produce a 50% increase in the clotting time relative to the clotting time in the absence of the inhibitor.
15. Ertl P. Polar surface area. In: Mannhold R, ed. *Molecular Drug Properties. Measurement and Prediction.* Weinheim: Wiley-VCH Verlag GmbH & Co. KGaA; 2008:111–126.
16. (a) Corte JR, Yang W, Fang T, et al. *Bioorg Med Chem Lett.* 2017;27:3833
    (b) Clark CG, Rossi KA, Corte JR, et al. *Bioorg Med Chem Lett.* 2019
    (c) Corte JR, Pinto DJP, Fang T, et al. *J Med Chem.* 2019. https://doi.org/10.1021/acs.jmedchem.9b01768.
17. Meanwell NA. *J Med Chem.* 2011;54(8):2529.
18. (a) Black WC, Bayly CI, Davis DE, et al. *Bioorg Med Chem Lett.* 2005;15:4741
    (b) Gauthier JY, Chauret N, Cromlish W, et al. *Bioorg Med Chem Lett.* 2008;18:923.
19. In addition, the –CH(CF_3)-NH-CH backbone angle is close to 120o observed with an amide, and the C-CF_3 bond is isopolar with the C=O. see Ref.[18a].
20. The absolute stereochemistry was assigned based on an X-ray co-crystal of 10 with FXIa. PDB deposition number for 10 is 5QTV.
21. Dunitz JD, Taylor R. *Chem Eur J.* 1997;3:89.
22. Replacing the NH with a CH_2 removed the H-bond interaction with Leu41.
23. PDB deposition number for 15 is 5QTW and 16 is 5QTX.
24. PDB deposition number for 24 is 5QTY.
25. He K, Qian M, Wong H, et al. *J Pharm Sci.* 2008;97:2568.
26. The corresponding 13-membered macrocycle 4 displayed a moderate clearance and showed no exposure.
27. Van Lierop BJ, Lummiss JAM, Fogg DE. Ring-closing metathesis. In: Grela K, ed. Olefin metathesis: theory and practice. Hoboken, New Jersey: John Wiley & Sons Inc. ; 2014:85–152.
28. Gong Y, Kato K. *J Fluor Chem.* 2004;125:767.
29. Chen Y, Dias HVR, Lovely CJ. *Tetrahedron Lett.* 2003;44:1379.